



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df019-2017-android-only
Operator:	Floris
Analysis Started:	2026-06-18 15:09:58
Analysis Completed:	2026-06-18 15:10:23
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df019\2017\controller-only-android
Output Directory:	C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df019\2017\dfdof_output\ft-df019-2017-android-only
Evidence Sources:	1
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df019_android.001	controller_android	physical	4.9 GB	2026-06-18 15:10:14

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\test_mavic_pro\df019\2017\controller-only-android

Cryptographic Hashes

df019_android.001	controller_android	physical
SHA-256:	dbc2abd14a23481723dc2536488369e74386ea035bce5d07ea2c12edd68f68bd	
SHA-1:	1fd0d0faeeef17017aca5c7915548a1dee07fca4	

Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p1 - controller android]: mmls: no partition table found in df019_android.001; treating as direct partition image
2. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for df019_android.001
3. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for df019_android.001
4. [p2 - controller ios]: source evidence not found
5. [p3 - controller android - drone logs]: no artefacts found
6. [p3 - controller ios]: source evidence not found
7. [p3 - drone sd]: source evidence not found
8. [p3 - drone flight storage]: source evidence not found
9. [p4 - controller ios]: source evidence not found
10. [p4 - drone sd]: source evidence not found
11. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
exiftool	3	3	[OK]
fls	1	1	[OK]
icat	76	76	[OK]
mmls	1	0	[X] Failed
parser_android	1	1	[OK]
txtlogtocsvtool	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	1 / 4
Artefact Categories With Data	6 / 7
Flights With Primary Correlation	0 / 1
Tools Succeeded	5 / 6

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	dji.go.v4	[K] Controller-only	1
Drone Serial Number	08RDD9100100KG	[K] Controller-only	2
Drone Model	Mavic Pro	[K] Controller-only	1
DJI Application Version	4.1.5	[K] Controller-only	1
Account E-Mail	droneforensics@vtoinc.com	[K] Controller-only	1
Product Type	13	[K] Controller-only	1
Last Country Code	US	[K] Controller-only	1
Aircraft Model Code	wm220	[K] Controller-only	1
Controller Serial Number	08TLDAQ00U034Q	[K] Controller-only	1
Last Known Latitude	39.96122700084717	[K] Controller-only	1
Last Known Longitude	-106.21635130627755	[K] Controller-only	1
Account UID	879825329191059456	[K] Controller-only	1
Account Nickname	dronefo...	[K] Controller-only	1

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: 4d81173e0f846fab43a5d858961f48ecb7853839551e5513da02accaabf8991d
Flight Record Name: DJIFlightRecord_2017-08-29__12-05-27_.csv
Start: 2017-08-29T18:05:27.929000+00:00
End: 2017-08-29T18:11:36.736000+00:00
Duration: 368.8 s

Flight Metrics

Peak Height: 29.8 m
Distance Travelled: 80.41 m
Video Recording: Yes
Photo Capture: No
Photos Taken: 0
Recording Duration: 179.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

1. EngineStart
2. AutoTakeoff
3. GPS_Atti
4. GoHome
5. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2017-08-29 18:05:27	ctrl_and:fr (v4.1.5)	Log started	SN: 08RDD9100100KG; App: 4.1.5; Mavic Pro
2017-08-29 18:05:27	ctrl_and:fr (v4.1.5)	Motor turned on	on
2017-08-29 18:05:29	ctrl_and:fr (v4.1.5)	Record mode changed	No
2017-08-29 18:08:36	ctrl_and:fr (v4.1.5)	Record mode changed	Starting
2017-08-29 18:10:48	ctrl_and:fr (v4.1.5)	Reached peak height	29.8 m
2017-08-29 18:11:35	ctrl_and:fr (v4.1.5)	Record mode changed	Stop
2017-08-29 18:11:36	ctrl_and:fr (v4.1.5)	Log ended	Dist home: 0.9 m; Height: 1.8 m; Rec: 179.0 s

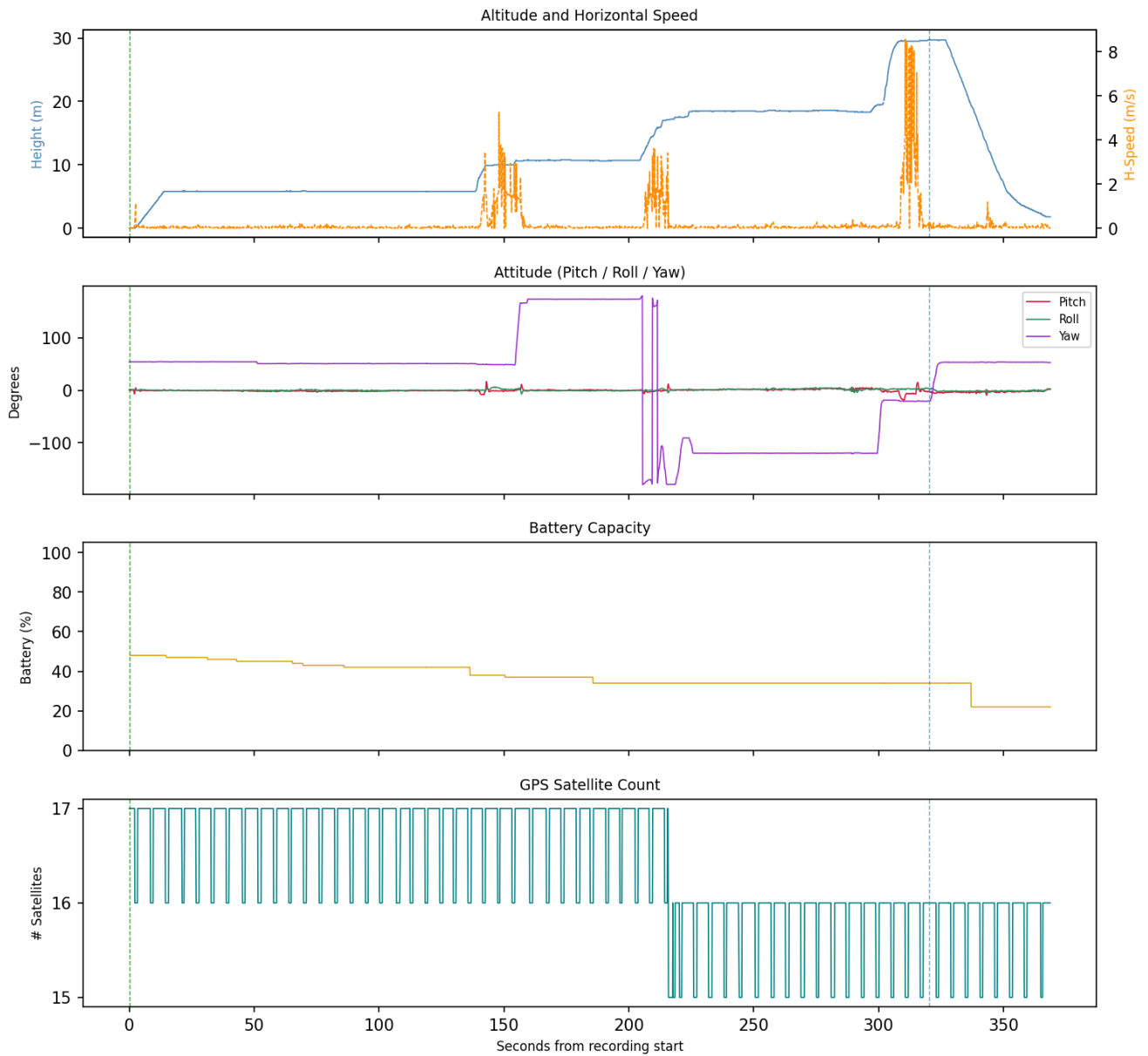
Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9611, lon -106.2163)
p5:1fe31d19fc31c3f92189767e9d33845ae531f5600d240795c34b311c2f27259f	
controller_android: flight_logs	Log messages match (12 entries, format: heading_log)
p5:440669680f9686525a0f7bebc808fe36e26680c76da0484ffdf119cb6f52f330	

Flight Record - Sensor Telemetry

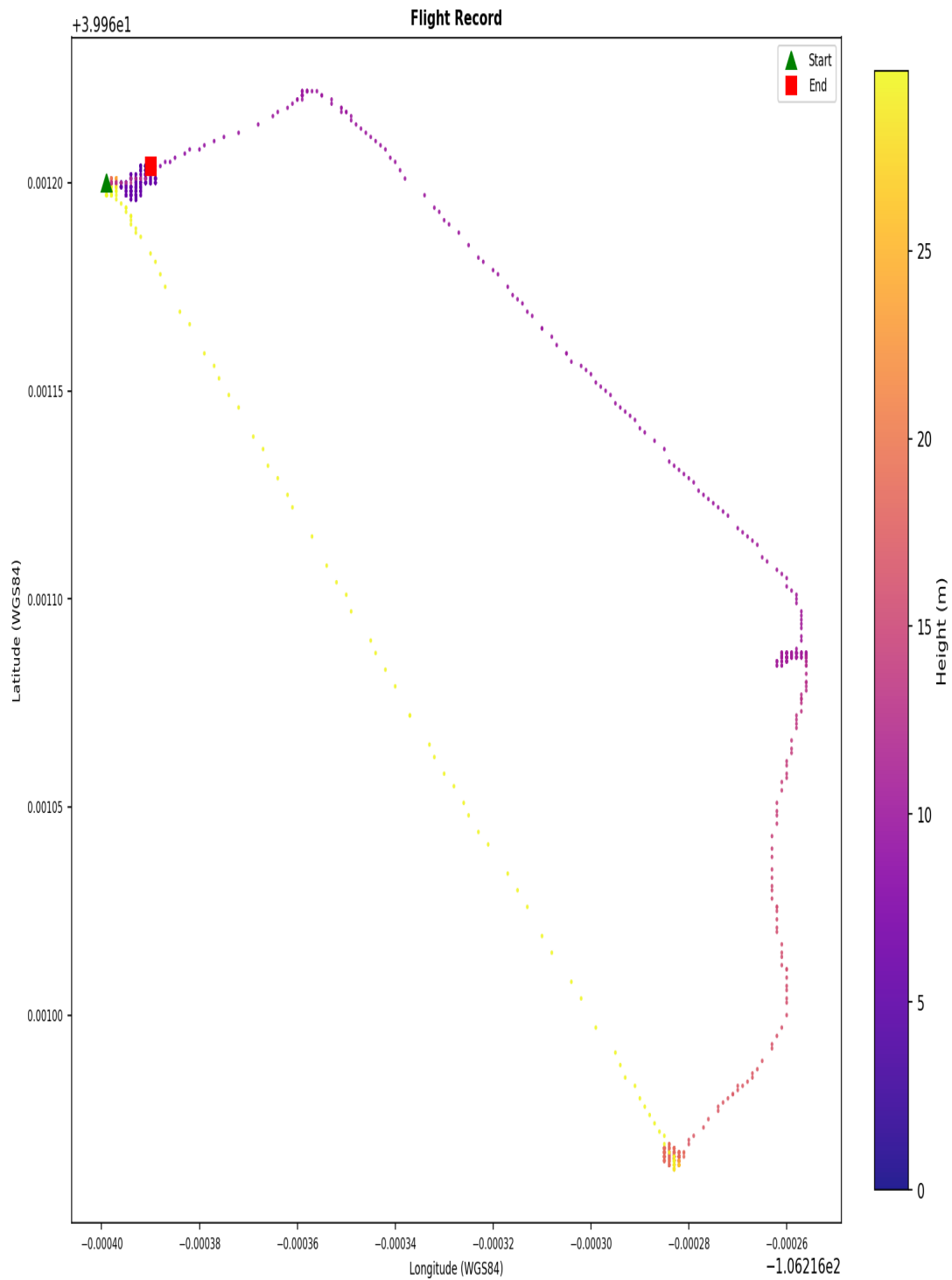
Flight Record Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_01



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	Yes
controller_ios	No
drone_sd	No
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_android	1	6	0	63	1	2	2

Data Quality Notes per Category

Flight Logs (63 files):

- 1 file(s) unreadable (binary)
- formats: 8 bracketed_log, 2 heading_log, 52 unknown

Flight Records (1 files):

- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Images (2 files):

- 2 file(s) missing (normalisable) EXIF timestamp
- 2 file(s) missing EXIF GPS coordinates

Videos (2 files):

- 1 file(s) contain(s) EXIF GPS coordinates
- 1 file(s) missing (normalisable) EXIF timestamp
- 1 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Flight_logs: 1 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: log files present in the following format(s): 8 bracketed_log, 2 heading_log, 52 unknown.
- Flight_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

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Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Flight Logs

26-08-2017-08RDD9100100KG

- format: heading_log | entries: 4

29-08-2017-08RDD9100100KG

- format: heading_log | entries: 12

log-2017-08-25.txt

- format: unknown | entries: 0

log-2017-08-25_1.txt

- format: unknown | entries: 0

log-2017-08-25_10.txt

- format: bracketed_log | entries: 1

log-2017-08-25_11.txt

- format: unknown | entries: 0

log-2017-08-25_12.txt

- format: unknown | entries: 0

log-2017-08-25_13.txt

- format: unknown | entries: 0

log-2017-08-25_14.txt

- format: unknown | entries: 0

log-2017-08-25_2.txt

- format: unknown | entries: 0

log-2017-08-25_3.txt

- format: bracketed_log | entries: 8

log-2017-08-25_4.txt

- format: unknown | entries: 0

log-2017-08-25_5.txt

- format: unknown | entries: 0

log-2017-08-25_6.txt
- format: unknown | entries: 0

log-2017-08-25_7.txt
- format: unknown | entries: 0

log-2017-08-25_8.txt
- format: unknown | entries: 0

log-2017-08-25_9.txt
- format: unknown | entries: 0

log-2017-08-26.txt
- format: unknown | entries: 0

log-2017-08-26_1.txt
- format: bracketed_log | entries: 1

log-2017-08-26_10.txt
- format: unknown | entries: 0

log-2017-08-26_11.txt
- format: unknown | entries: 0

log-2017-08-26_12.txt
- format: unknown | entries: 0

log-2017-08-26_13.txt
- format: unknown | entries: 0

log-2017-08-26_14.txt
- format: bracketed_log | entries: 1

log-2017-08-26_15.txt
- format: unknown | entries: 0

log-2017-08-26_2.txt
- format: unknown | entries: 0

log-2017-08-26_3.txt
- format: bracketed_log | entries: 6

log-2017-08-26_4.txt
- format: unknown | entries: 0

log-2017-08-26_5.txt
- format: unknown | entries: 0

log-2017-08-26_6.txt
- format: unknown | entries: 0

log-2017-08-26_7.txt
- format: unknown | entries: 0

log-2017-08-26_8.txt
- format: unknown | entries: 0

log-2017-08-26_9.txt
- format: unknown | entries: 0

log-2017-08-28.txt
- format: unknown | entries: 0

log-2017-08-28_10.txt
- format: unknown | entries: 0

log-2017-08-28_11.txt
- format: unknown | entries: 0

log-2017-08-28_12.txt
- format: unknown | entries: 0

log-2017-08-28_13.txt
- format: unknown | entries: 0

log-2017-08-28_14.txt

- format: bracketed_log | entries: 1

log-2017-08-28_15.txt

- format: unknown | entries: 0

log-2017-08-28_2.txt

- format: unknown | entries: 0

log-2017-08-28_3.txt

- format: bracketed_log | entries: 2

log-2017-08-28_4.txt

- format: unknown | entries: 0

log-2017-08-28_5.txt

- format: unknown | entries: 0

log-2017-08-28_6.txt

- format: unknown | entries: 0

log-2017-08-28_7.txt

- format: unknown | entries: 0

log-2017-08-28_8.txt

- format: unknown | entries: 0

log-2017-08-28_9.txt

- format: unknown | entries: 0

log-2017-08-29.txt

- format: unknown | entries: 0

log-2017-08-29_1.txt

- format: unknown | entries: 0

log-2017-08-29_2.txt

- format: unknown | entries: 0

log-2017-08-29_3.txt

- format: unknown | entries: 0

log-2017-08-29_4.txt

- format: unknown | entries: 0

log-2017-08-29_5.txt

- format: unknown | entries: 0

log-2017-08-29_6.txt

- format: unknown | entries: 0

log-2017-08-30.txt

- format: unknown | entries: 0

log-2017-08-30_1.txt

- format: unknown | entries: 0

log-2017-08-30_2.txt

- format: unknown | entries: 0

log-2017-08-30_3.txt

- format: bracketed_log | entries: 2

log-2017-08-30_4.txt

- format: unknown | entries: 0

log-2017-08-30_5.txt

- format: unknown | entries: 0

log-2017-08-30_6.txt

- format: unknown | entries: 0

Flight Records

norm_DJIFlightRecord_2017-08-29_12-05-27_.csv

- contains_no_value: 55 column(s) - APP_SER_WARN.warn, CENTER_BATTERY.connStatus, CENTER_BATTERY.errorType, CENTER_BATTERY.isBatteryOnCharge, CENTER_BATTERY.isNeedStudy (+50 more)

- contains_constant_value: 156 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_WARN.warn, CAMERA_INFO.cameraType (+151 more)

Images

thumb_32847c6fd0253b8c_1504030094000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

thumb_bbd8124c581d263e_1504028952000.jpg

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos

2017_08_29_12_08_36.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no controller device name found - Device and account information.
2. Note: no controller device found - Device and account information.
3. Note: no firmware version found - Device and account information.
4. Note: no device uuid found - Device and account information.
5. 1 flight record(s) could not be correlated to a drone log (flight_01). Independent corroboration is absent for these flight(s).
6. 6 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

dji.go.v4.xml	Account Data	controller_android	extract_physical
SHA-256:	8729652a658cb872873f49a9141a3ddaf00e6823275bbbbe433aad699dcfcabc		
SHA-1:	67ca11834efbb83b83e71e02547a735343dcafd6		

dafile.db	Databases	controller_android	extract_physical
SHA-256:	9e5b811e2fd45499450ff3bc2bd5051c468c58e6cd95d6f91cbd11ea8f78ba7b		
SHA-1:	34345884e99907a0fd151e006fb552e58b2b5766		

dji.db	Databases	controller_android	extract_physical
SHA-256:	4bb068ef59f5e9ebcdd9b9a60527d30ed8371313eba7684a67e414af29a34e2d		
SHA-1:	b20de411391a5020129a40adca4e7072054d6dc7		

flysafe_areas_djigo.db	Databases	controller_android	extract_physical
SHA-256:	d27dfcf41b9a2d6b2296f303702a201c1c333b3c2987a6465c0e5c5e83226601		
SHA-1:	cb48b850a53a541fdcdc13ffa66d677760ba685a		

flysafe_areas_djigo_1.db	Databases	controller_android	extract_physical
SHA-256:	296543bae3818032fd7d8763fab136d24daa2c4d9a67e39efcc85c9a5c10639f		
SHA-1:	77f4a2d11a090584d1c3b4d761de829da4238dd4		

flysafe_polygon_1860.db	Databases	controller_android	extract_physical
SHA-256:	bad78ebc14dcc7b63d2b807d0ced2d25d129d8bbc3d93285a6d986f06e0e8cd4		
SHA-1:	a2685bcff9b1ff7e82b4400f8e59b9ca3ff82ebb		

logdb.db	Databases	controller_android	extract_physical
SHA-256:	8c28563fe8db2d9ee5be5e86fa42a3a61cfe56bd01f8ebba3fd8b5c27aa8d72a		
SHA-1:	46d2de01107105544d8b6fe63c2153740eb6160e		

26-08-2017-08RDD9100100KG	Flight Logs	controller_android	extract_physical
SHA-256:	31ab88535821f70e0fcb5224dec9a9145a95a1728e69b0369e5bd4dcb4c98c48		
SHA-1:	fb830ca095bc79b3ba877d09c14df09e8ee7bbb1		

29-08-2017-08RDD9100100KG	Flight Logs	controller_android	extract_physical
SHA-256:	440669680f9686525a0f7bebc808fe36e26680c76da0484ffdf119cb6f52f330		
SHA-1:	2345994b0b808078e4eff523810b5b730edf0d97		

log-2017-08-25.txt	Flight Logs	controller_android	extract_physical
SHA-256:	fa38e790ed435dbceb71486b2bb1fc33694aaef0bd78ebddaef01074fd6c26c2		
SHA-1:	39de7d99aaa791becde6c6b8988367b7491493a7		

log-2017-08-25_1.txt	Flight Logs	controller_android	extract_physical
SHA-256:	b31c36853884ab72a7788e59881f07e640df2ba4d7181a2106b9078f9293bd33		
SHA-1:	c029c5dc9048a7af64a87ef9775a33d3100b368f		

log-2017-08-25_10.txt	Flight Logs	controller_android	extract_physical
SHA-256:	e29ca24a351688c134b89c4b384bc6247d5ce5449e89203ae835226bc9c10ce0		
SHA-1:	b29407cdc56eb9061081f4f81986d424494b3a4b		

log-2017-08-25_11.txt		Flight Logs	controller_android	extract_physical
SHA-256:	d5789fdafce04ac37dc51ef3ffbbcc538aaf26599884dc25d2969cd47fc94a75			
SHA-1:	46b50a15642dc9313c5ba7146c4751badcd9e45f			

log-2017-08-25_12.txt		Flight Logs	controller_android	extract_physical
SHA-256:	0cf67c46f0582429175aefaac5d4bb67fdd1f89d6a5fd57d06fe03ff3f5c5834			
SHA-1:	b7da27d6bc6000e7386ed0807721884f18385104			

log-2017-08-25_13.txt		Flight Logs	controller_android	extract_physical
SHA-256:	2b1b5cff36f3ca7fee9c46d5d8a7a7ed696e992e049ba89e2cc4e676d568caff			
SHA-1:	ea6d0af79de6957a4605abcd63f06363ab0f3fb9			

log-2017-08-25_14.txt		Flight Logs	controller_android	extract_physical
SHA-256:	b31d9f2a4c653187099d55b454e64458f32a938b1f8c725b9a6f080b5cd4f9ea			
SHA-1:	d4a339284675adc614295754c5be404232d4c9d7			

log-2017-08-25_2.txt		Flight Logs	controller_android	extract_physical
SHA-256:	a3daa5ac85ecc2260aa8de8cd5e69066dabe962ea3941d9d873b9a28b7c40996			
SHA-1:	84039cfe22202937d2dc72cfeb6a70e8ddf64568			

log-2017-08-25_3.txt		Flight Logs	controller_android	extract_physical
SHA-256:	1ca83b265ce29349027cbb3c502c7cd1e308e2b90bf6a0b67cac3bd5922afacf			
SHA-1:	89cb613748e9f01e5c600382fe403176e9fff3cb			

log-2017-08-25_4.txt		Flight Logs	controller_android	extract_physical
SHA-256:	b6a48c79bcd05b1d99a6eba0fcd17a0e0c1af0f935d46f336c1d27857c517e0e			
SHA-1:	3907ca2783e717bdb90a5a6ce5457264ed464772			

log-2017-08-25_5.txt		Flight Logs	controller_android	extract_physical
SHA-256:	4eb24c93d14f9cf023838949d4cf4be507fa53187490db21bd90e1b4a1be0470			
SHA-1:	055536083f790c5df1a076fffad691b29f836eff			

log-2017-08-25_6.txt		Flight Logs	controller_android	extract_physical
SHA-256:	cfb0d7cf61ccf563f4e53ad9f9958fde5bb5936b68c40b3ce4979d9dbf63ae87			
SHA-1:	2c9d96bcbf1940a04b8fff146cea0a17710c8e8f			

log-2017-08-25_7.txt		Flight Logs	controller_android	extract_physical
SHA-256:	0412df5055c7b2fb8aaf98bb41fd1f643b1b86eef482e7565826f6570aa3d721			
SHA-1:	fc521f4b5101d741af595e11808ca57ba06ae730			

log-2017-08-25_8.txt		Flight Logs	controller_android	extract_physical
SHA-256:	75e71403455be965f3ccbf426dbb6eb0cc71b0fead9c414e881feac34cfa9a23			
SHA-1:	bffd1906114a9664d49cc77ec67eb8133b060350			

log-2017-08-25_9.txt		Flight Logs	controller_android	extract_physical
SHA-256:	2e0899a50b908383d57f515749ee18b5b0aed74d5da02cf9e3aad624700b995			
SHA-1:	e9033e3e79e3e92c0f4d8b2c0f63a70058160e10			

log-2017-08-26.txt		Flight Logs	controller_android	extract_physical
SHA-256:	09087a19a310434153acd7a47d0ed15e033c37f760af17e57b219fd42d31d46c			
SHA-1:	dfea20ebfc5371845b4577e57ee9199bcfe735fb			

log-2017-08-26_1.txt	Flight Logs	controller_android	extract_physical
SHA-256:	962cca7e12f3e5db451a24e9c814ea515d0e01feb6f51c43b78efc6f9b415352		
SHA-1:	a37ae110ca51f5c227cb0e994b82823cd0299273		

log-2017-08-26_10.txt	Flight Logs	controller_android	extract_physical
SHA-256:	7591f3ebfb4ef747d151293d69aeddccbc5a415e96b9e41521cb490d20c3cd0		
SHA-1:	792c7e0561030d6f249ee87ef214daddad5a33ea		

log-2017-08-26_11.txt	Flight Logs	controller_android	extract_physical
SHA-256:	04f8bc7a5402733aa7cc6895f3cc69f108f7f31cc355039f66697fe247211b44		
SHA-1:	fa20c01faaa19385f505c65599d27a6e6b63be0a		

log-2017-08-26_12.txt	Flight Logs	controller_android	extract_physical
SHA-256:	6218f2779e97f302c3f160dd3c6d45262580a3a71fb854a04ee623d08e42d074		
SHA-1:	7df7092f215364ac2896450b44da44f6fd5b5827		

log-2017-08-26_13.txt	Flight Logs	controller_android	extract_physical
SHA-256:	52d3edb57fb8b1134ee33a40980c74791853b979a17905b8c43af61e0a0cd21d		
SHA-1:	c7e7add440b4e2720be99f15f82e996cb5930c45		

log-2017-08-26_14.txt	Flight Logs	controller_android	extract_physical
SHA-256:	6374a0b978f03666eda82ab5440dab367b945c0b1aa5d4eb234a77e0c39b9fa7		
SHA-1:	32b457fc91eb364066cb1723f3c1fca21df71306		

log-2017-08-26_15.txt	Flight Logs	controller_android	extract_physical
SHA-256:	422fb5f0ccd9df855ce0fa60e409f7c0cd918610b4a842af54ff4570cb332ddc		
SHA-1:	0080ba4cda71b1335a57d0e9d0d93d1843e5047e		

log-2017-08-26_2.txt	Flight Logs	controller_android	extract_physical
SHA-256:	ac4a2cafbfb1ef6d6b68fc61ae0c05ba9cb34fe14d51d1e33af5d5d646430cf05		
SHA-1:	daa1f4cec48296aca8de7286bce63c9733b052a3		

log-2017-08-26_3.txt	Flight Logs	controller_android	extract_physical
SHA-256:	a3c9cb9dedefd7150bd7144882761d64808a08a42c43e7aa2915a082e1a681f1		
SHA-1:	792683cfc04061efb8ce970d32f75acf2c94608c		

log-2017-08-26_4.txt	Flight Logs	controller_android	extract_physical
SHA-256:	427fbcdb719d3931ec2b04417bd4c4a887b1e3d96bc77455392795ece19ff99		
SHA-1:	483114d208fa72e027d93ce9e8248a4e6f7417c9		

log-2017-08-26_5.txt	Flight Logs	controller_android	extract_physical
SHA-256:	b26294a5871e9de454e881991ab0317896b4a4216cf031844366c7ab7ec5f5bf		
SHA-1:	e8351d1418655711fdc06cfae718ad4e2da9e542		

log-2017-08-26_6.txt	Flight Logs	controller_android	extract_physical
SHA-256:	98a23c70cdcb972c09e3fb6344416f2004f857541e138a79364eaffd00173273		
SHA-1:	e99f650a2f45d1584485d54dcf32ce67abbd6ac5		

log-2017-08-26_7.txt	Flight Logs	controller_android	extract_physical
SHA-256:	1bb2dc04eb7f8a36397e47e82c1e56a98a194149217816a2e909f570fff28499		
SHA-1:	82dce87609f14e9ca39eaf89a5c175fe7288864		

log-2017-08-26_8.txt	Flight Logs	controller_android	extract_physical
SHA-256:	490140e2a29fe183cf6dfe5609885e773ae9cfea147e5841aa5abe069ade3d0b		
SHA-1:	50c7dcfb650f860ce26ecd0bdd6ece56992f0ae0		

log-2017-08-26_9.txt	Flight Logs	controller_android	extract_physical
SHA-256:	ba6814ef24e72db97d8adab2ae29627362c3d92f2b931ee2f54e31f881fe1344		
SHA-1:	757d5634e8185b510be1a61ed07f06685b67c187		

log-2017-08-28.txt	Flight Logs	controller_android	extract_physical
SHA-256:	3663e56830f3e3f5fba90a6f1b5c904e8cec11df742f483944672989699112aa		
SHA-1:	e35b697a69b8516088974ad33c6b6685e94672b4		

log-2017-08-28_1.txt	Flight Logs	controller_android	extract_physical
SHA-256:	f9267495d7af368c3d2cc8d4e4ba5a4fde4273e6ee1fcc5c3b9c564fa54f164c		
SHA-1:	b8c533394dc934cb0374d837d17ca0cafeb1c9bd		

log-2017-08-28_10.txt	Flight Logs	controller_android	extract_physical
SHA-256:	74a8697076de842d26031e8b58183d471887ff0ee839b6a8c32404629b0f6aa7		
SHA-1:	a1d9d1715330042312fcc438fd29d570d541a493		

log-2017-08-28_11.txt	Flight Logs	controller_android	extract_physical
SHA-256:	84fba68fab14c232a35dfdb86effcd078602dfd483856328bac5ca107d9f9865		
SHA-1:	180ea9ed1e01330656cfb7034e5c5ad5707bff9b		

log-2017-08-28_12.txt	Flight Logs	controller_android	extract_physical
SHA-256:	364c2d8e368ab1d3e4c30fa9f09414dee53813fe23439953ff30d2b32dd6c25b		
SHA-1:	28db49b1ef9156db992a33cad900b3ad84fd1e22		

log-2017-08-28_13.txt	Flight Logs	controller_android	extract_physical
SHA-256:	9140e12e1dd7e591217a638e85fc5620dd58a5ae754cc12b07380d3c6dd924a0		
SHA-1:	8209b3f98d5afdb1b6dcd738c6b3abbab74dbf29		

log-2017-08-28_14.txt	Flight Logs	controller_android	extract_physical
SHA-256:	ca666fc969001191f910e3680a965cbf84a43fdee6b59005492c64ead9d8a32f		
SHA-1:	9635f0bec741b7321eee40d1e4d9ed5a09e13a17		

log-2017-08-28_15.txt	Flight Logs	controller_android	extract_physical
SHA-256:	33527ea46073a90d57afb59a4ce220217e31e5308679a325d3a2078750a605e5		
SHA-1:	32089fd959cec272d99c681764c74c4b7013c16b		

log-2017-08-28_2.txt	Flight Logs	controller_android	extract_physical
SHA-256:	7f6778f86c1634a3a4f3db5c66d06bb9cb25bea5696732a6403d6582712d1169		
SHA-1:	9b7845718891ab43fcd890c64d69ac7511b78e09		

log-2017-08-28_3.txt	Flight Logs	controller_android	extract_physical
SHA-256:	e28cb3b3f3d1583bc8cf2d334ec78ccf6ea99ab9892abbce9084e9e49d858d81		
SHA-1:	f9314f48efad9be3066a385e203525815255c8bc		

log-2017-08-28_4.txt	Flight Logs	controller_android	extract_physical
SHA-256:	c20c88a9977936eba4cb9917db6d2dfbc79e19e5fd21e956c714becd70c48a89		
SHA-1:	c76135676a07cb987457284afba29be8048e0808		

log-2017-08-28_5.txt		Flight Logs	controller_android	extract_physical
SHA-256:	52371c5b5ee8574e4e8764423ea084ecd85093a85f4281254b014ed7ffe00044			
SHA-1:	ef745b12f9b6244921595777e4276d3cc9c9bd945			

log-2017-08-28_6.txt		Flight Logs	controller_android	extract_physical
SHA-256:	3c223c11a8aa22acb7506d8be7d18cea902fd34faf515214eb6dd59f454de044			
SHA-1:	34c084ffb452fdef1385f0373a5680a83a993b19			

log-2017-08-28_7.txt		Flight Logs	controller_android	extract_physical
SHA-256:	996f308448380cd000ad888c38b4dc5533780512bf4ddd9060ad8e3b82cb1075			
SHA-1:	fb53692404b2ac2bb23b8ba0de133ecc4bdf1ada			

log-2017-08-28_8.txt		Flight Logs	controller_android	extract_physical
SHA-256:	a1f8a07be24dcea285e3b9eae68e4fba1310a0371515ed7d115e7e0133c9a017			
SHA-1:	cefbab363cd90a24225b89da92791fac3207778b			

log-2017-08-28_9.txt		Flight Logs	controller_android	extract_physical
SHA-256:	366fa69e13f096864045b5aa859ee78c92a868907854f21503b1bd294201ef56			
SHA-1:	56b689af6c2230766583b024f9939da0ede144fa			

log-2017-08-29.txt		Flight Logs	controller_android	extract_physical
SHA-256:	cede10dd85029af5326c66719a908815ae721dd26ad29a18191e4c4e970a62d1			
SHA-1:	4cc2770bc18ec3a95a33a08c8c3bda0cbb3d1a99			

log-2017-08-29_1.txt		Flight Logs	controller_android	extract_physical
SHA-256:	419925e6bad35eba7cc59bbd598175480b9b83523cbaa3b43db952e7ef423ae5			
SHA-1:	3248d89235dce86c0964b784af737b94de1ec147			

log-2017-08-29_2.txt		Flight Logs	controller_android	extract_physical
SHA-256:	9d1352d9cf384263d091ef5434967f735c4705f7697ea1f19fb002004b44836d			
SHA-1:	4452587dbbeb928450a99a951deddb10f858c9c8			

log-2017-08-29_3.txt		Flight Logs	controller_android	extract_physical
SHA-256:	41d784708bf9192f0bfc173794efe6e695d50e91063f48757b230c01727ed625			
SHA-1:	a3ae2cc002ecd851a7aed68b188690c149c9b5ea			

log-2017-08-29_4.txt		Flight Logs	controller_android	extract_physical
SHA-256:	87ed69dea6386ec3cc6b6cd7059bc7afe8444a6db49b91fab9fb23defba93c24			
SHA-1:	fc465cc7d2098e1532d58ac43d747785e2a2d424			

log-2017-08-29_5.txt		Flight Logs	controller_android	extract_physical
SHA-256:	aa9d2f65f3f0203c082bcce62f178856bd2c4ef8eebec8afa86356c3d7125a87			
SHA-1:	d74bd9eec6f1797c07c645a7169bc192760b7bc0			

log-2017-08-29_6.txt		Flight Logs	controller_android	extract_physical
SHA-256:	56c17c9701ecb888e29c4c054ddf107d43da916dca3e5fe5407cc1cd2c112fef			
SHA-1:	4fa72b6538e87256be4e84f61f0ac8480291e731			

log-2017-08-30.txt		Flight Logs	controller_android	extract_physical
SHA-256:	12b5702f91e8f65bb3b15981ecec1e9f86ae56ab4a2059e472db871643b7759a			
SHA-1:	76488ef9108bc3f795871fe4ef8cd7a7326d58e1			

log-2017-08-30_1.txt		Flight Logs	controller_android	extract_physical
SHA-256:	4dfab67f52c26e0a54f407e7a437861d0a7fe738ab605eec1770bc6835f407f			
SHA-1:	1fd2ba52c1191b0cadd91455f4f5df8cf0a056fb			

log-2017-08-30_2.txt		Flight Logs	controller_android	extract_physical
SHA-256:	07af021128be64df3355c79a9fc6b6848d2c843af07bb9c4ede8dc1b88efffa75			
SHA-1:	e1483122a6bb1a7efdafc9e321402b1aa01aa1bb			

log-2017-08-30_3.txt		Flight Logs	controller_android	extract_physical
SHA-256:	d1c03a9ac384e31a2a3abbc418b2b8f7bc37eef297aa2579ad371f904a1a629c			
SHA-1:	285db60fffd5908de466e44cded1202de02bc868			

log-2017-08-30_4.txt		Flight Logs	controller_android	extract_physical
SHA-256:	61920118338af884edaf04c0c3ac33327f8c2556303caf91db533596a94f56ba			
SHA-1:	2131632f00d92fde7b9c44fdb7f55994bd057a67			

log-2017-08-30_5.txt		Flight Logs	controller_android	extract_physical
SHA-256:	1980c05dff0e679bf51593f03298e2a282beafb29d85cd6b77a73152fea1acdf			
SHA-1:	a7032eed98994ac073d8144fc0f4591a01cf452a			

log-2017-08-30_6.txt		Flight Logs	controller_android	extract_physical
SHA-256:	5b29ea284cfce6b71185d7c367372f66acd8d3275fa24cd5d133fd2993d6b18f			
SHA-1:	071fd5e6153a403b10c5d5f3a9ded11c54128934			

DJIFlightRecord_2017-08-29_12-05-27_.txt		Flight Records	controller_android	extract_physical
SHA-256:	cbe116774d30a094cc14d8843e9d64fb6ff0a36f7a9b3caa78e42fd517005dd0			
SHA-1:	4293b2d83da36d6963232c8b14337e9e7a8997b1			

thumb_32847c6fd0253b8c_1504030094000.jpg		Images	controller_android	extract_physical
SHA-256:	d276c695bab4544212e3a28681e551f6353aba7aa8f211261a86d5189d82511b			
SHA-1:	54af65ab2add083d9d8fc73a2a98031d440115fe			

thumb_bbd8124c581d263e_1504028952000.jpg		Images	controller_android	extract_physical
SHA-256:	72dbc2f9db9bd470fe6b19554348d4d70ef0dd7e1c8dd897729d15a4ab22d76d			
SHA-1:	ef10928f4c74175f0ad50c7d977b1f3787b4e9b5			

2017_08_29_12_08_36.info		Videos	controller_android	extract_physical
SHA-256:	1fe31d19fc31c3f92189767e9d33845ae531f5600d240795c34b311c2f27259f			
SHA-1:	53a030fa29ef4690c05b98e5bb4f322cd65c4cf7			

2017_08_29_12_08_36.mp4		Videos	controller_android	extract_physical
SHA-256:	976d1778e862648c719863a2088a13666f7d095d1adb2a25b574f18864066ae4			
SHA-1:	18d9a54629acc0e5f47f8533a91886cbf6800fef			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Android

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	1	1	0
Databases	6	0	0
Flight Logs	63	0	0
Flight Records	1	1	1
Images	2	2	0
Videos	2	2	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-18 15:09:58 mmls vThe Sleuth Kit ver 4.15.0 rc=1

cmd: mmls.exe df019_android.001

2026-06-18 15:09:58 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p df019_android.001

2026-06-18 15:10:14 parser_android v- rc=n/a

cmd: df019_android.001 C:\Users\Floris\Documents\dfdo_output\ft-df019-2017-android-only\p2_image_parsing\controller_android_parsed
out: controller_android_parsed

2026-06-18 15:10:14 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 470

out: packages.list

2026-06-18 15:10:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1828

out: dji.go.v4.xml

2026-06-18 15:10:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 66260

out: flysafe_areas_djigo.db

2026-06-18 15:10:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 66267

out: flysafe_polygon_1860.db

2026-06-18 15:10:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 650

out: dji.db

2026-06-18 15:10:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 946

out: logdb.db

2026-06-18 15:10:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1833

out: flysafe_areas_djigo_1.db

2026-06-18 15:10:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1842

out: dafile.db

2026-06-18 15:10:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 632

out: log-2017-08-25.txt

2026-06-18 15:10:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1603

out: log-2017-08-26.txt

2026-06-18 15:10:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2153

out: log-2017-08-28.txt

2026-06-18 15:10:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2195

out: log-2017-08-29.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2026

out: log-2017-08-30.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 999

out: log-2017-08-25_1.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 311

out: log-2017-08-26_1.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2155

out: log-2017-08-28_1.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2197

out: log-2017-08-29_1.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1987

out: log-2017-08-30_1.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1003

out: log-2017-08-25_2.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1823

out: log-2017-08-26_2.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2157

out: log-2017-08-28_2.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2092

out: log-2017-08-29_2.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2200

out: log-2017-08-30_2.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1005

out: log-2017-08-25_3.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1824

out: log-2017-08-26_3.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2158

out: log-2017-08-28_3.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2201

out: log-2017-08-29_3.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2205

out: log-2017-08-30_3.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1011

out: log-2017-08-25_4.txt

2026-06-18 15:10:16 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1822

out: log-2017-08-26_4.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2156

out: log-2017-08-28_4.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2173

out: log-2017-08-29_4.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2196

out: log-2017-08-30_4.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1410

out: log-2017-08-25_5.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 931

out: log-2017-08-26_5.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1632

out: log-2017-08-28_5.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2154

out: log-2017-08-29_5.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 503

out: log-2017-08-30_5.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1424

out: log-2017-08-25_6.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1249

out: log-2017-08-26_6.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1841

out: log-2017-08-28_6.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1436

out: log-2017-08-25_7.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1605

out: log-2017-08-26_7.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2162

out: log-2017-08-28_7.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1438

out: log-2017-08-25_8.txt

2026-06-18 15:10:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1832

out: log-2017-08-26_8.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2099

out: log-2017-08-28_8.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1440

out: log-2017-08-25_9.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1592

out: log-2017-08-26_9.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1864

out: log-2017-08-28_9.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 506

out: log-2017-08-25_10.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1826

out: log-2017-08-26_10.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1963

out: log-2017-08-28_10.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1789

out: log-2017-08-25_11.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1250

out: log-2017-08-26_11.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2129

out: log-2017-08-28_11.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1816

out: log-2017-08-25_12.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1837

out: log-2017-08-26_12.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2164

out: log-2017-08-28_12.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1817

out: log-2017-08-25_13.txt

2026-06-18 15:10:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1838

out: log-2017-08-26_13.txt

2026-06-18 15:10:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2130

out: log-2017-08-28_13.txt

2026-06-18 15:10:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 66456

out: log-2017-08-26_14.txt

2026-06-18 15:10:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 66319

out: log-2017-08-28_14.txt

2026-06-18 15:10:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1269

out: log-2017-08-25_14.txt

2026-06-18 15:10:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1618

out: log-2017-08-26_15.txt

2026-06-18 15:10:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 994

out: log-2017-08-28_15.txt

2026-06-18 15:10:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1434

out: log-2017-08-29_6.txt

2026-06-18 15:10:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1487

out: log-2017-08-30_6.txt

2026-06-18 15:10:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 66439

out: 26-08-2017-08RDD9100100KG

2026-06-18 15:10:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 66704

out: 29-08-2017-08RDD9100100KG

2026-06-18 15:10:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 162

out: DJIFlightRecord_2017-08-29__12-05-27_.txt

2026-06-18 15:10:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 1839

out: thumb_bbd8124c581d263e_1504028952000.jpg

2026-06-18 15:10:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 2152

out: thumb_32847c6fd0253b8c_1504030094000.jpg

2026-06-18 15:10:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 66515

out: 2017_08_29_12_08_36.mp4

2026-06-18 15:10:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df019_android.001 66714

out: 2017_08_29_12_08_36.info

2026-06-18 15:10:21 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2017-08-29__12-05-27_.txt DJIFlightRecord_2017-08-29__12-05-27_.csv

out: DJIFlightRecord_2017-08-29__12-05-27_.csv

2026-06-18 15:10:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_32847c6fd0253b8c_1504030094000.jpg

out: thumb_32847c6fd0253b8c_1504030094000_exif.json

2026-06-18 15:10:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n thumb_bbd8124c581d263e_1504028952000.jpg

out: thumb_bbd8124c581d263e_1504028952000_exif.json

2026-06-18 15:10:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_08_29_12_08_36.mp4

out: 2017_08_29_12_08_36_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.